

Super Intelligence and General Intelligence

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Abstract—Artificial general intelligence (AGI) and superintelligence are frequently treated as successive points on a single scale of cognitive performance. This paper argues that they represent distinct capabilities. Building on intelligence as information-dependent decision capability and AGI as integrated multimodal agency, we define general intelligence by an agent’s capacity to coordinate perception, reasoning, memory, tools, decisions, and actions across domains. Superintelligence requires an additional epistemic transition: the ability to expand the frontier of knowledge beyond both what humanity knows and the questions humanity has already learned to ask. We distinguish three related processes. Knowledge fusion establishes consequential relations among previously separated bodies of knowledge. Knowledge discovery identifies previously unknown phenomena, mechanisms, regularities, or formal objects. Knowledge invention occurs when a machine originates a novel epistemic structure that becomes consequential, independently validated, and incorporable into collective knowledge. We formalize this distinction through the Knowledge-Invention Proposition and propose a staged evaluation framework based on frontier verification, unknown-unknown origination, causal attribution, independent validation, consequentiality, and repeatability. The framework distinguishes machine assistance, machine discovery, and validated machine knowledge invention while addressing contamination, human specification, epistemic opacity, and dual-use risk. Finally, we maintain that consciousness, general intelligence, and superintelligence are analytically independent. Superintelligence is therefore not defined by the quantity of knowledge stored or performance on known tasks, but by the repeatable capacity of a generally intelligent machine to originate validated knowledge beyond the human epistemic frontier.

Index Terms—Super Intelligence, AGI, General Intelligence, Artificial Consciousness, Machine Consciousness

I. INTRODUCTION

Intelligence, consciousness, general intelligence, and superintelligence are frequently placed on a single ascending scale. This treatment obscures the different questions addressed by these concepts. Intelligence concerns a system’s capacity to use information in selecting among possible actions, states, or outputs. Consciousness concerns the organization and, in its phenomenal sense, the subjective character of experience. General intelligence concerns the breadth, transferability, and adaptive organization of agency. Superintelligence, we argue, concerns a further capability: moving beyond the collective human frontier by inventing validated knowledge that humanity did not previously possess or recognize as an object of inquiry.

Our earlier work defined intelligence at its minimal foundation as information-dependent selection while distinguishing intelligence from both functional and phenomenal consciousness [1]. On this account, intelligence is graded rather than binary. A simple conditional selection may instantiate an elementary decision structure, whereas advanced systems can coordinate memory, inference, planning, metacognition, and action. Increasing sophistication can therefore produce more organized intelligence without proving the existence or character of subjective experience.

Consciousness should likewise not be restricted in advance to one biological, humanlike, or computational form. We previously argued that phenomenal, observable, and functional claims about consciousness must be separated and that language can serve as a unifying interface among specialized artificial minds [2]. Language is not merely an input–output modality under this account. It can provide a shared representational medium through which perceptual, reasoning, memory, and supervisory systems exchange information. Such coordination may enable increasingly integrated levels of functional artificial consciousness without, by itself, establishing phenomenal experience.

These distinctions become essential when considering artificial general intelligence (AGI). Our account of AGI as integrated multimodal agency treats general intelligence and consciousness as related but non-identical dimensions [3]. Neither multimodality nor consciousness alone supplies general intelligence. AGI requires the persistent coordination of perception, memory, reasoning, objectives, planning, metacognition, and adaptive action across unfamiliar tasks, domains, and environments. This view is consistent with accounts that relate intelligence to an agent’s ability to achieve goals across environments rather than to performance on one fixed task [4] and with broader treatments of AGI as adaptable, domain-general intelligence rather than a collection of isolated task systems [5].

Future work will introduce *AGI-Former*, a neural architecture intended to operationalize this conception of integrated multimodal agency. The present paper does not specify that architecture. It instead addresses the conceptual problem that follows from it: even if a machine realizes AGI, what additional capability would make it superintelligent?

A. From General Intelligence to Superintelligence

General intelligence and superintelligence are sometimes treated as interchangeable descriptions of an increasingly capable system. They instead identify different functional achievements. AGI denotes integrated, transferable, and adaptive agency across tasks, domains, modalities, and environments. Superintelligence denotes capability beyond the collective human frontier: the capacity of a generally intelligent machine to originate validated knowledge that humanity did not previously possess or recognize as an object of inquiry. This retains the established comparative meaning of superintelligence as capability exceeding human intelligence [6], while specifying the epistemic capability that distinguishes it from AGI in the present framework.

An AGI could therefore remain bounded by existing human knowledge. It might reason across disciplines, operate tools, adapt plans, and solve difficult problems while deriving its objectives and conceptual boundaries from questions already posed by humans. These abilities would establish general agency, but they would not alone demonstrate superintelligence. The transition occurs when the machine ceases to be only a general user of known science and becomes an originator of new science.

This distinction can be expressed directly:

$$\begin{aligned} \text{AGI}(x) &\not\Rightarrow \text{SI}(x), \\ \text{SI}(x) &\Rightarrow \text{AGI}(x). \end{aligned} \quad (1)$$

The additional condition is not greater speed, scale, memory, or performance within a predetermined task. It is machine-originated knowledge invention. To state this condition precisely, the distinction between known unknowns and unknown unknowns must first be established.

B. Known Unknowns and Unknown Unknowns

Scientific research commonly begins with a *known unknown*: a question, target, mechanism, or missing value that researchers have already identified but not yet resolved. A machine may answer such a question more quickly or accurately than human researchers and thereby make a major scientific contribution. The machine is operating beyond prior human performance, but the epistemic destination was already represented within human inquiry.

An *unknown unknown* is stronger. It is a consequential question, relation, phenomenon, explanatory possibility, or field of inquiry that humanity has neither resolved nor recognized as a target. Unknown unknowns are not simply entries missing from a database. Before they are exposed, humans do not know that a particular question should be asked, that two bodies of knowledge should be connected, that an observed anomaly signifies a new mechanism, or that a new conceptual object is possible.

Let $\mathcal{K}_{H,t}$ denote the collective body of justified human knowledge at time t , and let $\mathcal{Q}_{H,t}$ denote the set of questions, hypotheses, and research targets explicitly recognized

by the human scientific community. For a candidate epistemic contribution κ ,

$$\begin{aligned} \kappa \in \mathcal{Q}_{H,t} \wedge \kappa \notin \mathcal{K}_{H,t} &\implies \kappa \text{ is a known unknown,} \\ \kappa \notin \mathcal{Q}_{H,t} \wedge \kappa \notin \mathcal{K}_{H,t} &\implies \kappa \text{ is an unknown unknown.} \end{aligned} \quad (2)$$

The second relation cannot be established solely by observing that a generated sentence is absent from a training corpus. Semantic novelty, scientific significance, and epistemic validity are different properties. A system must identify or construct the previously unrecognized object of inquiry, explain its relation to existing knowledge, and produce evidence through which the claim can be independently assessed.

Existing AI-for-science results demonstrate important progress without yet collapsing this distinction. AI has recovered symbolic relations from data [7], guided mathematicians toward new conjectures and theorems [8], predicted protein structures at high accuracy [9], and expanded computational catalogues of candidate stable materials [10]. These achievements show that machine learning can contribute to discovery at the scientific frontier. They also remain embedded in research problems, objectives, data, and validation practices established by human investigators. Under the present framework, their scientific importance is not in question; what remains to be established is whether a machine can itself expose an unknown unknown and originate the knowledge through which it becomes understood.

C. Knowledge Invention

We use *knowledge invention* for the machine-originated production of a new epistemic object that lies beyond the collective human frontier and becomes justified through appropriate validation. The term includes the invention of new explanatory theories, conceptual structures, scientific methods, architectures, or research directions. It does not imply that natural laws are arbitrarily created. A machine may discover a previously unknown phenomenon or relation, but the representation, explanation, and generalizable body of knowledge constructed from that discovery constitute an invented epistemic contribution.

Knowledge fusion and knowledge discovery are principal routes toward knowledge invention. Knowledge fusion connects bodies of evidence, concepts, models, or disciplines whose consequential relation was not previously recognized. Knowledge discovery identifies a previously unknown phenomenon, mechanism, regularity, entity, or law. Either process may produce knowledge invention when the machine converts its result into a novel, consequential, and validated addition to collective knowledge:

$$\begin{aligned} \text{Fusion}_x &\longrightarrow \kappa_F, \\ \text{Discovery}_x &\longrightarrow \kappa_D, \\ \text{Validate}(\kappa_F \vee \kappa_D) = 1 &\longrightarrow \text{KI}_x(\kappa), \end{aligned} \quad (3)$$

where $\text{KI}_x(\kappa)$ denotes knowledge invention attributable to machine x . The arrows represent possible epistemic pathways

rather than a claim that every fusion or discovery yields invention. Most recombinations are not consequential, many hypotheses are false, and some discoveries answer questions already posed by humans. Knowledge invention therefore requires a stronger conjunction:

$$\begin{aligned} \text{KI}_x(\kappa) \iff & \kappa \notin \mathcal{K}_{H,t} \\ & \wedge \kappa \notin \mathcal{Q}_{H,t} \\ & \wedge \text{Originate}_x(\kappa) \\ & \wedge \text{Consequential}(\kappa) \\ & \wedge \text{Validate}(\kappa) = 1. \end{aligned} \quad (4)$$

The origination condition does not require isolation from humans. Scientific work is commonly distributed across people, instruments, institutions, and computational systems. It requires that the machine make the causally essential move that exposes or constructs the unknown unknown, rather than merely execute a fully specified human procedure. The consequentiality condition excludes arbitrary novelty. The validation condition may be satisfied through controlled experiment, mathematical proof, reproducible computation, successful engineering realization, or another method appropriate to the domain. Until validation occurs, the output remains a candidate invention or hypothesis rather than knowledge.

Recent systems already integrate language models with literature search, code, planning, laboratory interfaces, and experimental execution [11], while autonomous laboratories combine computation, machine learning, robotics, and empirical feedback in closed experimental loops [12]. Reviews of AI-enabled science similarly show that learning systems now participate across hypothesis generation, prediction, experimental design, and analysis [13]. These systems provide architectural and empirical precursors to machine knowledge invention. They do not by their existence satisfy Equation (4), because scientific autonomy within a human-specified objective is not identical to originating an unrecognized object of inquiry.

D. The Defining Transition to Superintelligence

Under the framework proposed here, AGI is necessary but not sufficient for superintelligence. An AGI may transfer knowledge, adapt to unfamiliar environments, and pursue objectives across domains while remaining bounded by the human knowledge and research targets available to it. Superintelligence begins when general intelligence can originate and validate knowledge beyond that collective frontier:

$$\begin{aligned} \text{SI}(x) \iff & \text{AGI}(x) \\ & \wedge \exists \kappa \text{ KI}_x(\kappa) \\ & \wedge \mathcal{K}_{H,t+1} = \mathcal{K}_{H,t} \cup \{\kappa\}, \\ \text{SI}(x) \Rightarrow & \text{AGI}(x), \\ \text{AGI}(x) \not\Rightarrow & \text{SI}(x). \end{aligned} \quad (5)$$

The central thesis of this paper is therefore that *superintelligence is general intelligence capable of originating validated knowledge beyond the collective human frontier, including the*

identification and resolution of unknown unknowns. Knowledge fusion and knowledge discovery are not three parallel requirements alongside knowledge invention. They are principal generative routes through which knowledge invention may occur.

This definition does not equate superintelligence with omniscience, infallibility, consciousness, model size, or universal superiority on every task. It makes a narrower and testable claim: the defining evidence for superintelligence is a machine-originated epistemic contribution that humanity did not previously possess or explicitly seek, that materially extends human understanding, and that survives independent validation. The remainder of the paper distinguishes this capability from general intelligence, examines knowledge fusion and discovery as routes to invention, and develops criteria for attributing knowledge invention to an artificial system without mistaking recombination, hallucination, or unverified novelty for science.

II. FROM GENERAL INTELLIGENCE TO SUPERINTELLIGENCE

Section I proposed that superintelligence begins when a generally intelligent machine originates validated knowledge beyond the collective human frontier. This section develops that distinction more precisely. The transition from AGI to superintelligence is not defined by a larger model, a longer context window, access to more tools, or improved performance within an existing task. It is defined by a change in the machine's epistemic role: from a system that operates across established knowledge and human-specified problems to a system that can expose, formulate, and resolve problems that humanity had not recognized.

A. The Functional Boundary of General Intelligence

General intelligence concerns the organization of agency. An AGI must transfer knowledge across domains, recompose existing capabilities for unfamiliar problems, maintain goals over time, revise plans after failure, and coordinate perception, memory, reasoning, metacognition, and action [3]. Formal definitions of intelligence likewise emphasize the capacity to achieve goals across a wide range of environments [4], while broader AGI accounts distinguish general adaptability from competence on isolated tasks [5].

Let $\mathcal{E}$ denote environments, $\mathcal{T}$ tasks, and $\mathcal{M}$ modalities. For agent x , let T_x , R_x , and A_x denote transfer, recomposition, and adaptation. The functional scope of general agency can be represented as

$$\begin{aligned} \mathcal{G}(x) &= \Phi_x(T_x, R_x, A_x), \\ \Phi_x : \mathcal{E} \times \mathcal{T} \times \mathcal{M} &\longrightarrow \text{coherent goal-directed behavior}, \end{aligned} \quad (6)$$

where Φ_x denotes the persistent organization through which these capacities inform and constrain one another. No single component is sufficient. Transfer without adaptation may fail when prior assumptions break; recomposition without coherent goals may produce locally useful but globally inconsistent

actions; and broad task exposure may reflect memorized coverage rather than general agency.

Generality also cannot be inferred from the mere presence of many components. A framework may contain language, vision, memory, search, code execution, and robotic interfaces while relying on a human to select tools, reconcile their outputs, detect failure, and preserve the objective. Framework capacity, realized agency, and observed performance must therefore remain distinct:

$$\begin{aligned} \text{Cap}(F) &\not\equiv \text{Realize}(x, F), \\ \text{Realize}(x, F) &\not\equiv \text{Perf}(x, \tau), \end{aligned} \quad (7)$$

where F is a computational framework and τ is an evaluated task. An AGI realizes coordinated general agency; it need not be omniscient, infallible, or superior on every task. Its defining property is the ability to reorganize its available capacities in pursuit of objectives across sufficiently varied and unfamiliar conditions.

B. The Epistemic Boundary of General Intelligence

Nothing in the preceding definition requires AGI to create knowledge beyond humanity. A generally intelligent machine could possess extensive human knowledge, apply it across disciplines, generate plans, conduct analyses, and solve previously unsolved but already formulated problems. Such a system could be transformative while remaining epistemically bounded by the concepts, questions, and objectives inherited from human inquiry. Reviews of AI-enabled scientific discovery already distinguish multiple roles for machines across hypothesis generation, modelling, experimental design, and analysis [13]. That breadth of participation motivates a more precise question of attribution: whether the machine merely operates within the human research agenda or originates an epistemic object beyond it.

Let $\mathcal{K}_{H,t}$ be collective justified human knowledge at time t , and let $\mathcal{Q}_{H,t}$ be the set of questions, hypotheses, and research targets recognized by humans. An AGI operating within the human epistemic boundary may be represented by

$$\begin{aligned} x : (\mathcal{K}_{H,t}, \mathcal{Q}_{H,t}, \mathcal{O}_t) &\longrightarrow (\mathcal{Y}_t, \mathcal{A}_t), \\ \mathcal{Q}_{x,t} &\subseteq \mathcal{Q}_{H,t}, \end{aligned} \quad (8)$$

where $\mathcal{O}_t$ denotes available observations, $\mathcal{Y}_t$ generated conclusions or predictions, and $\mathcal{A}_t$ actions. The subset relation expresses the limiting case in which the machine's research agenda remains contained within questions humans have already recognized. The machine may produce an answer absent from $\mathcal{K}_{H,t}$ and thereby resolve a known unknown, but it does not yet originate the object of inquiry itself.

This distinction is visible in current AI-assisted science. Machine learning has helped recover symbolic laws from experimental data [7], [14], guided mathematicians toward conjectures and theorems [8], predicted protein structures [9], and proposed large numbers of candidate stable materials [10]. Language models connected to tools and laboratory automation can design, plan, and execute chemistry experiments

[11], while autonomous laboratories can perform closed-loop materials synthesis [12]. Earlier robot-scientist systems likewise generated and experimentally tested functional-genomics hypotheses [15] and demonstrated end-to-end automation of iterative scientific investigation [16]. These are consequential advances at the scientific frontier. They also begin from domains, objectives, datasets, experimental systems, and validation conditions established by human researchers. Their results demonstrate increasingly powerful scientific instruments and agents, not by themselves the invention of an unknown unknown.

C. Autonomous Problem Formation

The first additional requirement for superintelligence is autonomous problem formation. The machine must be capable not only of answering a question but of identifying that a consequential question exists. This may occur when the system detects an unexplained regularity, recognizes a contradiction across disciplines, constructs a new conceptual variable, identifies an assumption shared by existing theories, or notices that available observations support a previously unrepresented phenomenon. Work on automated identification of latent state variables from high-dimensional observations illustrates progress toward machines constructing representations that were not specified directly by researchers [17]. Such representation discovery is an important precursor, although it does not alone establish autonomous problem formation in the stronger sense proposed here.

Let q_x be a machine-originated research question. Autonomous problem formation requires

$$\begin{aligned} q_x &\notin \mathcal{Q}_{H,t}, & q_x &\notin \mathcal{K}_{H,t}, \\ \text{Ground}(q_x, \mathcal{K}_{H,t}, \mathcal{O}_t) &= 1, & \text{Consequential}(q_x) &= 1. \end{aligned} \quad (9)$$

The grounding condition prevents arbitrary question generation from being mistaken for intelligence. A language model can generate indefinitely many unusual questions through stochastic variation. Most will be trivial, ill-posed, physically impossible, semantically incoherent, or disconnected from evidence. A superintelligent problem formulation must arise from a defensible relation among observations, established knowledge, unresolved tensions, or formally represented possibilities. The consequentiality condition further requires that resolving the question would materially alter understanding, explanation, prediction, or technological capability.

Autonomous problem formation is therefore more demanding than novelty. It requires epistemic judgment: the system must distinguish an unasked question that matters from an unasked question that is merely possible. This capacity is the point at which an unknown unknown becomes an identified object of inquiry.

D. Causal Origination of Knowledge

Scientific results are commonly produced by distributed systems containing researchers, datasets, instruments, software, models, and institutions. Requiring a superintelligent machine

to work without human participation would therefore impose a standard not applied to human scientists. Nevertheless, attribution cannot be granted merely because AI appears somewhere in the workflow. The machine must make the causally essential epistemic contribution.

Let W be a scientific workflow and let κ be its resulting knowledge claim. We define machine origination counterfactually:

$$\begin{aligned} \text{Originate}_x(\kappa \mid W) = 1 &\iff \text{Produce}(W, \kappa) = 1 \\ &\quad \wedge \text{Produce}(W \setminus x, \kappa) = 0 \\ &\quad \wedge \text{Specified}_H(\kappa) = 0. \end{aligned} \quad (10)$$

The first condition requires that the workflow produce the contribution. The second requires that removing the machine's relevant epistemic operation would prevent that contribution from being produced through the same workflow. The third excludes cases in which humans fully specified the conclusion and used the machine only to calculate, retrieve, format, or experimentally execute it. This is a conceptual causal test rather than a demand for literal repetition of every discovery process.

Machine origination can occur through knowledge fusion or knowledge discovery. In fusion, the essential contribution may be recognizing that previously separated theories, data, or disciplines instantiate a common structure. In discovery, it may be detecting a new entity, mechanism, regularity, or law. Neither pathway is sufficient until its output is converted into an epistemic object that can be understood, assessed, and incorporated into knowledge. Prior systems for symbolic-law recovery and automated hypothesis testing demonstrate that portions of this cycle can be computationally realized [14], [15]; the unresolved issue is whether the machine can originate the consequential object of inquiry rather than search within a human-defined hypothesis space.

E. The Transition Criterion

The transition from AGI to superintelligence requires a complete epistemic cycle. The machine must form or expose the previously unrecognized problem, originate a candidate explanation or solution, connect that contribution to existing knowledge, and enable domain-appropriate validation. Let κ_x denote the resulting machine-originated contribution. Then

$$\begin{aligned} \text{KI}_x(\kappa_x) &\iff \text{FormProblem}_x(q_x) \\ &\quad \wedge \text{Originate}_x(\kappa_x \mid W) \\ &\quad \wedge \text{Explain}_x(\kappa_x, \mathcal{K}_{H,t}) \\ &\quad \wedge \text{Validate}(\kappa_x) = 1 \\ &\quad \wedge \text{Consequential}(\kappa_x) = 1. \end{aligned} \quad (11)$$

Knowledge invention is not established at the moment a machine produces a novel output. Before validation, the output is a candidate hypothesis, theory, method, or invention. Validation may require controlled experiment, formal proof, reproducible computation, engineering realization, convergent

evidence, or another procedure accepted by the relevant discipline. Independent human or machine verification may participate in this process, but the validation standard must not be defined solely by the originating system. This insistence on domain-appropriate validation follows the broader AI-for-science literature, in which prediction, hypothesis generation, experimentation, and analysis form an interdependent discovery process rather than interchangeable evidence of knowledge [13].

The relationship between AGI and superintelligence can consequently be stated as

$$\begin{aligned} \text{SI}(x) &\iff \text{AGI}(x) \\ &\quad \wedge \exists q_x, \kappa_x : q_x \notin \mathcal{Q}_{H,t} \\ &\quad \wedge \text{KI}_x(\kappa_x) \\ &\quad \wedge \mathcal{K}_{H,t+1} = \mathcal{K}_{H,t} \cup \{\kappa_x\}. \end{aligned} \quad (12)$$

Equation (12) preserves the established meaning of superintelligence as capability beyond humans [6] while replacing an unspecified performance comparison with an epistemic criterion. The machine goes beyond humans not merely by doing a known task better, but by making a validated addition to collective knowledge whose object of inquiry humanity had not recognized.

This criterion also clarifies why the present framework does not define a separate category of domain superintelligence. A specialized scientific model may be indispensable to a discovery, but specialization does not establish general intelligence. Under the terminology developed here, superintelligence is a mode of general intelligence. Specialized systems are scientific instruments or components of a discovery system unless they also realize the general agency and machine-originated knowledge invention required by Equation (12).

Knowledge fusion and knowledge discovery now require separate analysis because they describe the principal routes by which a machine may cross the human epistemic boundary. The following sections examine how each route can produce knowledge invention, how their outputs differ from retrieval and recombination, and what evidence would justify attributing the resulting knowledge to the machine.

III. KNOWLEDGE FUSION AS A PATHWAY TO KNOWLEDGE INVENTION

Knowledge is distributed across disciplines, institutions, datasets, terminologies, and historical periods. Two scientific communities may study related structures without sharing concepts, cite separate literatures while describing interacting mechanisms, or possess results that become consequential only when interpreted together. This fragmentation creates a space in which the components of a new insight may already be publicly available even though the insight itself is absent from collective knowledge. Knowledge fusion is the process by which an intelligent system identifies and constructs such a missing relation.

The possibility of deriving new hypotheses from disconnected literatures has a substantial history. Swanson described

undiscovered public knowledge through the example of separately published biomedical findings whose latent connection suggested a previously unrecognized hypothesis [18]. Later work on literature-based discovery generalized this idea into computational methods for identifying implicit relations across scientific corpora. Studies of scientific innovation likewise show that atypical combinations of prior work are associated with unusually high impact [19], while research that bridges distant regions of a knowledge network is riskier but can yield greater rewards [20]. These findings motivate knowledge fusion, but the definition developed here imposes requirements beyond bibliographic novelty.

A. From Combination to Fusion

Let $\mathcal{K}_1, \dots, \mathcal{K}_n$ denote bodies of justified knowledge associated with distinct domains, communities, or representational systems. Their aggregation is the union

$$\mathcal{K}_\cup = \bigcup_{i=1}^n \mathcal{K}_i. \quad (13)$$

Aggregation increases availability but does not create a new epistemic relation. A search engine, database, or language model may retrieve elements from several domains and place them in one response without changing what is known. Knowledge fusion requires an additional contribution κ_F that is not explicitly contained in any source domain or in their simple union:

$$\begin{aligned} \mathcal{F}_x(\mathcal{K}_1, \dots, \mathcal{K}_n) &\longrightarrow \kappa_F, \\ \kappa_F &\notin \mathcal{K}_i \quad \forall i, \\ \kappa_F &\notin \mathcal{K}_\cup. \end{aligned} \quad (14)$$

Here, $\mathcal{F}_x$ is the fusion process realized by machine x , and κ_F is a candidate relation, explanation, hypothesis, or conceptual structure. The final condition means that κ_F cannot be recovered as an explicit claim already present in the aggregated sources. Its components may be present, but their consequential connection is not.

This distinction separates five increasingly demanding operations. Retrieval locates an existing item. Aggregation places items together. Synthesis organizes existing claims into a coherent account. Recombination rearranges known elements into a configuration that may be novel in form. Fusion constructs a relation that changes what can be explained, predicted, tested, or created. Only the last operation is a candidate pathway to knowledge invention.

B. Latent Relations Across Knowledge Domains

A latent relation exists when justified claims in separated knowledge domains jointly support an implication not represented within any participating domain. The simplest structure is the open-discovery pattern

$$\begin{aligned} A &\longrightarrow B, \\ B &\longrightarrow C, \\ A &\overset{?}{\longrightarrow} C, \end{aligned} \quad (15)$$

where the solid relations are established and the dashed relation is an unrecognized candidate inferred through the shared intermediary B . This structure captures the foundation of classical literature-based discovery [18], but knowledge fusion need not be limited to a three-node chain. It may connect causal graphs, mathematical structures, experimental anomalies, measurement techniques, or explanatory models across many domains.

The importance of a fusion does not follow from distance alone. Two remote concepts may be unrelated, and two adjacent concepts may support a profound inference. Empirical studies of scientific production indicate that successful high-impact work often balances conventional foundations with atypical combinations [19]. A fusion system must therefore preserve enough established structure to support inference while exploring relations that prevailing disciplinary organization makes unlikely to be examined.

Let $d(\mathcal{K}_i, \mathcal{K}_j)$ measure conceptual or bibliographic distance, and let $J(\kappa_F)$ represent the justification available for a candidate fusion. Then neither novelty nor distance alone is sufficient:

$$\begin{aligned} d(\mathcal{K}_i, \mathcal{K}_j) &\gg 0 \not\Rightarrow J(\kappa_F) > 0, \\ J(\kappa_F) > 0 &\not\Rightarrow \text{Consequential}(\kappa_F) = 1. \end{aligned} \quad (16)$$

The first relation rejects arbitrary cross-domain analogy. The second rejects valid but trivial connections. Knowledge fusion requires both defensible support and consequential epistemic change.

C. Fusion as the Exposure of an Unknown Unknown

Knowledge fusion becomes relevant to superintelligence when it reveals not only a new answer but a previously unrecognized object of inquiry. Suppose $\mathcal{K}_a$ and $\mathcal{K}_b$ contain individually coherent models, yet their joint implications produce a contradiction, unexplained regularity, or missing causal variable. A machine may use that tension to formulate a new question q_F :

$$\begin{aligned} \mathcal{F}_x(\mathcal{K}_a, \mathcal{K}_b) &\longrightarrow q_F, \\ q_F &\notin \mathcal{Q}_{H,t}, \\ \text{Ground}(q_F, \mathcal{K}_a \cup \mathcal{K}_b) &= 1. \end{aligned} \quad (17)$$

The question is an unknown unknown before fusion because humanity has not represented it within the recognized research agenda. Fusion changes that state. It exposes the question, provides its initial justification, and creates a target for discovery or theory construction. Computational work that identifies previously unspecified state variables from observations provides a partial example of machines constructing new representational candidates [17]. The stronger fusion criterion additionally requires that the machine recognize the cross-domain significance of the constructed object.

Scientific search is shaped by human incentives and by the organization of existing knowledge. Network analyses show that researchers preferentially pursue established relationships even when riskier, more innovative searches have greater

potential value [20]. Computational systems can search differently. Models of collective discovery suggest that strategic selection of experiments can accelerate the exploration of a scientific knowledge network [21]. A fusion-oriented machine could extend this principle by selecting not only experiments but also distant knowledge regions whose joint structure merits investigation.

D. Fusion-Driven Knowledge Invention

A candidate fusion becomes knowledge invention only after it is developed into a consequential epistemic contribution and validated. Let $\text{FKI}_x(\kappa_F)$ denote fusion-driven knowledge invention by machine x . Then

$$\begin{aligned} \text{FKI}_x(\kappa_F) \iff & \text{Fuse}_x(\mathcal{K}_1, \dots, \mathcal{K}_n) \\ & \wedge \kappa_F \notin \mathcal{K}_{H,t} \\ & \wedge \text{Originate}_x(\kappa_F \mid W) \\ & \wedge \text{Consequential}(\kappa_F) \\ & \wedge \text{Validate}(\kappa_F) = 1. \end{aligned} \quad (18)$$

Equation (18) excludes a system that merely retrieves a previously published interdisciplinary connection. It also excludes a system that proposes a novel connection without evidence, repeats a human specified synthesis, or generates an interesting analogy with no explanatory or predictive consequence. The machine must originate the essential relation and that relation must survive independent, domain-appropriate evaluation.

Validation depends on the type of fused contribution. A causal hypothesis may require controlled intervention; a mathematical bridge may require proof; a unified explanatory model may require superior prediction across the source domains; and a proposed technological principle may require successful realization. AI-enabled science already distributes hypothesis generation, modelling, experimental planning, and analysis across interacting systems [13]. Fusion-driven knowledge invention adds the requirement that the cross-domain relation itself originate from the machine rather than be specified by the human workflow.

E. Failure Modes and Evidentiary Safeguards

Knowledge fusion is especially vulnerable to outputs that appear profound because they connect distant concepts. At least six failure modes must be distinguished from genuine fusion. First, *lexical overlap* mistakes shared terminology for shared mechanism. Second, *analogy inflation* treats a structural resemblance as evidence of causal identity. Third, *correlation substitution* presents statistical association as explanation. Fourth, *ontology mismatch* combines concepts whose meanings or measurement assumptions are incompatible. Fifth, *hidden retrieval* reproduces a connection present in the training corpus or literature while presenting it as machine-originated. Sixth, *novelty without consequence* generates a defensible but scientifically unimportant relation.

For a candidate κ_F , a minimum evidentiary record should therefore contain

$$\mathcal{R}(\kappa_F) = \{ \text{Sources, Provenance, Derivation, NoveltySearch, Predictions, Validation} \}. \quad (19)$$

Source and provenance records identify the knowledge used. Derivation records show how the new relation follows from or transforms those inputs. Novelty search tests whether the relation already exists in human knowledge. Predictions state what should be observed if the fusion is correct. Validation records preserve the independent evidence used to accept, reject, or revise the claim. This record does not eliminate epistemic uncertainty, but it makes the attribution of machine-originated fusion auditable.

Knowledge fusion thus reveals what separated bodies of knowledge imply when connected. It can expose an unknown unknown and provide the conceptual structure from which knowledge invention develops. Knowledge discovery, the subject of the next section, addresses a different pathway: identifying a phenomenon, mechanism, entity, or regularity that existing knowledge does not yet adequately represent, even after its known components have been connected.

IV. KNOWLEDGE DISCOVERY AS A PATHWAY TO KNOWLEDGE INVENTION

Knowledge fusion derives a consequential relation by connecting bodies of knowledge that were previously separated. Knowledge discovery begins from a different epistemic condition: some feature of reality is not adequately represented in existing knowledge, even after known information has been assembled. The missing object may be a phenomenon, entity, mechanism, variable, regularity, law, material, algorithm, or causal relation. A discovery system must therefore do more than reorganize descriptions. It must establish that reality contains a structure that collective human knowledge did not previously capture.

Machine-assisted discovery is already an active scientific program. Symbolic regression has recovered compact physical relations from experimental data [7], [14]; sparse identification has extracted governing dynamical equations from measurements [22]; and representation-learning systems have proposed latent state variables from raw observations [17]. In biology and chemistry, machine learning has contributed to protein-structure prediction [9], antibiotic identification [23], materials prediction [10], and closed-loop experimental synthesis [12]. These achievements establish that machines can participate in multiple stages of discovery. The present framework asks when that participation becomes machine-originated knowledge invention.

A. Prediction, Detection, and Discovery

Prediction is not identical to discovery. A system may accurately map inputs to outputs without identifying a new entity, mechanism, or explanatory structure. Detection is also insufficient. A detector may locate an anomaly or classify an unfamiliar object while leaving its significance unexplained. Optimization can improve an established design without

changing the relevant knowledge. These operations become components of discovery only when they support a justified claim about something previously unknown.

Let o denote an observation, p a prediction, a a detected anomaly, and κ_D a candidate discovery. The non-equivalence is

$$\begin{aligned} \text{Predict}_x(p \mid o) &\not\Rightarrow \text{Discover}_x(\kappa_D), \\ \text{Detect}_x(a \mid o) &\not\Rightarrow \text{Discover}_x(\kappa_D), \\ \text{Optimize}_x(y) &\not\Rightarrow \text{Discover}_x(\kappa_D). \end{aligned} \quad (20)$$

A prediction may contribute evidence; an anomaly may motivate a hypothesis; and optimization may expose an unexpected solution. Discovery nevertheless requires an epistemic transition from an output to a validated claim about the world or about a formally defined possibility space.

This distinction is illustrated by data-driven law discovery. Sparse identification methods can produce parsimonious governing equations from noisy measurements [22], while symbolic-regression systems can recover interpretable relations rather than only point predictions [7], [14]. Their explanatory representations move closer to discovery because they state a general relation that can be tested outside the observations used to construct it. Even then, recovery of a known law is rediscovery or methodological validation, not an extension of the human knowledge frontier.

B. Objects of Machine Discovery

Machine discovery may target several types of epistemic object. A *phenomenon* is an observed behavior or effect not adequately represented by existing theory. An *entity* is a previously unknown object, class, material, biological structure, or computational construction. A *mechanism* explains how interacting processes produce an effect. A *variable* supplies a representation required to describe a system. A *regularity* expresses a stable relation among observations, and a *law* provides a generalizable mathematical or causal structure. An *algorithm* is a previously unknown procedure that produces a formally or empirically verifiable result.

These objects require different validation standards. A candidate material must be computationally and ultimately experimentally assessed; a natural law must predict across conditions; a causal mechanism must survive intervention; and an algorithm may be established through proof, exhaustive verification, or implementation benchmarks. Machine-generated algorithmic results demonstrate the importance of this distinction. AlphaTensor discovered provably correct matrix-multiplication algorithms that improved on known procedures [24], and AlphaDev produced new sorting routines that were validated and incorporated into a widely used standard library [25]. These are genuine algorithmic discoveries within human-formulated problem spaces. Under the present account, they demonstrate a major component of knowledge invention but not yet the autonomous exposure of an unknown unknown.

Let Ω denote the relevant space of possible objects and $\mathcal{R}_{H,t}(\Omega)$ the region represented in collective human knowledge at time t . A candidate machine discovery occupies

$$\begin{aligned} \kappa_D &\in \Omega, \\ \kappa_D &\notin \mathcal{R}_{H,t}(\Omega), \\ \text{Represent}_x(\kappa_D) &= 1. \end{aligned} \quad (21)$$

The representation condition requires the machine to distinguish the object from noise and express it in a form that supports examination. It does not yet establish that the representation is true, important, or attributable to the machine.

C. The Discovery Cycle

Knowledge discovery requires a connected cycle rather than a single model output. The cycle begins with observations and proceeds through anomaly or structure detection, hypothesis formation, explanatory modeling, discriminating tests, and validation:

$$\begin{aligned} \mathcal{O}_t &\longrightarrow \mathcal{A}_x \\ &\longrightarrow \mathcal{H}_x \\ &\longrightarrow \mathcal{M}_x \\ &\longrightarrow \mathcal{T}_x \\ &\longrightarrow \mathcal{V}(\kappa_D), \end{aligned} \quad (22)$$

where $\mathcal{O}_t$ contains observations, $\mathcal{A}_x$ contains machine-identified anomalies or structures, $\mathcal{H}_x$ contains hypotheses, $\mathcal{M}_x$ contains explanatory models, $\mathcal{T}_x$ contains discriminating tests, and $\mathcal{V}(\kappa_D)$ is the validation result.

The cycle need not be linear. Failed tests may revise the representation, generate alternative hypotheses, expose measurement error, or cause the system to seek new observations. Autonomous laboratories demonstrate how prediction, planning, experimentation, interpretation, and revision can be connected in a closed loop [11], [12]. Earlier robot scientists likewise showed that machine-generated hypotheses could be selected and tested experimentally [15], [16]. These systems provide architectural precedents for the discovery cycle, although their research domains and objectives were established by humans.

For discovery beyond the human frontier, the machine must exercise epistemic control across the cycle. It must determine which observation is anomalous, which hypothesis is explanatory rather than merely predictive, which test can distinguish competing accounts, and how the result changes the candidate knowledge claim. A pipeline whose scientific choices are supplied entirely by humans may automate discovery work without realizing machine discovery in this strong sense.

D. From Anomaly to Explanation

Anomalies are important because they may signal that an existing model is incomplete, but most anomalies are not discoveries. They may result from noise, instrument error, data leakage, distribution shift, implementation defects, or unrepresented confounders. A discovery system must discriminate among these possibilities before attributing an anomaly to a new feature of reality.

Let a be an anomaly under current model M_H , and let $\{h_1, \dots, h_n\}$ be competing explanations. The system should

select a test τ^* according to its expected ability to discriminate among them:

$$\tau^* = \arg \max_{\tau \in \mathcal{T}} \text{Disc}(\tau; h_1, \dots, h_n), \quad (23)$$

$$\text{Disc}(\tau; h_1, \dots, h_n) > \delta,$$

where Disc measures the expected discriminating value and δ is a domain-appropriate threshold. Strategic experiment selection has been proposed as a means of accelerating collective scientific discovery [21]. Within the present framework, the stronger requirement is that the machine select tests in service of a hypothesis it causally originated about a previously unrecognized object.

Explanation also protects against mistaking correlation for discovery. A model may predict an outcome from a proxy variable while failing under intervention or distributional change. A candidate discovery should therefore identify conditions under which its explanation would fail. Falsifiable consequences, counterfactual behavior, and out-of-distribution predictions provide stronger evidence than retrospective fit alone.

E. Discovery-Driven Knowledge Invention

A validated discovery becomes knowledge invention when the machine originates both the discovery object and the epistemic structure through which it enters collective knowledge. Let $\text{DKI}_x(\kappa_D)$ denote discovery-driven knowledge invention. Then

$$\begin{aligned} \text{DKI}_x(\kappa_D) \iff & \kappa_D \notin \mathcal{K}_{H,t} \\ & \wedge q_D \notin \mathcal{Q}_{H,t} \\ & \wedge \text{Discover}_x(\kappa_D) \\ & \wedge \text{Originate}_x(\kappa_D \mid W) \\ & \wedge \text{Explain}_x(\kappa_D, \mathcal{K}_{H,t}) \\ & \wedge \text{Validate}(\kappa_D) = 1 \\ & \wedge \text{Consequential}(\kappa_D) = 1. \end{aligned} \quad (24)$$

The first two conditions place the object and its motivating question beyond current human knowledge and inquiry. The discovery condition requires the machine to identify the object from evidence or formal search. The origination condition assigns the causally essential epistemic step to the machine. The explanation condition relates the object to existing knowledge, while validation and consequentiality prevent false, trivial, or uninterpretable novelty from being classified as knowledge invention.

This definition separates three levels of achievement:

$$\begin{aligned} \text{CandidateDiscovery} \subset \text{ValidatedDiscovery} \\ \subset \text{DiscoveryDrivenKnowledgeInvention}. \end{aligned} \quad (25)$$

A candidate discovery is an unvalidated representation or hypothesis. A validated discovery establishes that the object or relation is real within the relevant standard. Discovery-driven knowledge invention additionally requires machine origination

beyond a human-recognized research target and integration of the result into an explanatory epistemic structure.

F. Failure Modes and Validation Requirements

Machine discovery is vulnerable to systematic errors. *Data artifacts* can masquerade as phenomena. *Overfitting* can produce laws that describe observations but fail elsewhere. *Proxy discovery* can identify a predictive correlate instead of a mechanism. *Search-space confinement* can make an output appear unprecedented even though the system explored only a human-specified formal space. *Measurement dependence* can create entities that disappear under a different instrument or preprocessing method. *Rediscovery* can reproduce knowledge absent from the system's immediate context but present elsewhere in the literature. Finally, *validation capture* occurs when the originating model also defines and evaluates the only test of its claim.

For candidate discovery κ_D , the evidentiary package should contain

$$\begin{aligned} \mathcal{E}(\kappa_D) = \{ & \text{ObservationProvenance, NoveltySearch,} \\ & \text{AlternativeHypotheses, DiscriminatingTests,} \\ & \text{Replication, BoundaryConditions, Validation} \}. \end{aligned} \quad (26)$$

Observation provenance preserves the origin and transformations of the data. Novelty search tests whether the discovery already exists in collective knowledge. Alternative hypotheses prevent premature commitment. Discriminating tests separate the proposed account from competitors. Replication assesses whether the result survives independent implementation, measurement, or experimentation. Boundary conditions specify where the claim should and should not hold. Validation records the evidence and standards used to accept the contribution.

Knowledge discovery therefore identifies what reality or a formal possibility space contains beyond existing representation. It becomes a pathway to superintelligence only when a generally intelligent machine originates an unknown object of inquiry, constructs and tests an explanation, and produces a validated, consequential addition to collective human knowledge. Knowledge fusion and knowledge discovery can also interact: fusion may reveal the question that discovery answers, while discovery may supply the missing object that makes a deeper fusion possible. The next section integrates these pathways under the paper's central concept of knowledge invention.

V. KNOWLEDGE INVENTION: THE DEFINING CAPABILITY OF SUPERINTELLIGENCE

Sections III and IV identified two pathways beyond existing knowledge. Fusion constructs a consequential relation among previously separated bodies of knowledge. Discovery establishes a previously unknown phenomenon, entity, mechanism, variable, regularity, law, or formal object. Neither pathway alone fully specifies the transition to superintelligence. A fused relation may remain an untested hypothesis, and a discovered

phenomenon may remain unexplained. Knowledge invention occurs when a generally intelligent machine originates the epistemic structure through which such a result becomes consequential, validated, and incorporable into collective knowledge.

The term *invention* is used deliberately. Natural phenomena are not invented by the system, but theories, explanatory structures, representations, methods, algorithms, architectures, and research programs are constructed. Machine knowledge invention may therefore begin with discovery, fusion, or their interaction, but it culminates in a new epistemic object that did not previously exist within either collective human knowledge or the recognized human research agenda.

A. Novelty, Usefulness, and Epistemic Validity

Research on creativity commonly treats originality and effectiveness as joint requirements [26]. Work on artificial creativity likewise distinguishes novel combination, exploration within a conceptual space, and transformation of that space [27]. These distinctions are relevant to knowledge invention but remain insufficient for scientific attribution. An idea may be original and useful without being true, and a true statement may be trivial, previously known, or accidentally generated.

Let κ be a candidate epistemic contribution. Knowledge invention requires at least four properties:

$$\begin{aligned} N(\kappa) &= 1, \\ C(\kappa) &= 1, \\ V(\kappa) &= 1, \\ O_x(\kappa) &= 1, \end{aligned} \quad (27)$$

where N denotes novelty relative to collective human knowledge, C consequentiality, V domain-appropriate validation, and O_x causal origination by machine x . The conjunction is essential. Novelty without consequence is arbitrary variation; consequence without validity is an attractive error; validity without novelty is rediscovery; and novelty, consequence, and validity without machine origination describe a human invention assisted by a machine rather than machine knowledge invention.

The standard is therefore stronger than computational creativity. A system may generate a novel analogy, design, proof sketch, molecule, or theory candidate and still fail to invent knowledge. The output becomes knowledge only after its claims survive evaluation independent of the originating process.

B. Discovery, Fusion, and Invention

Fusion and discovery are neither mutually exclusive nor guaranteed to produce invention. Fusion may expose a contradiction that becomes a new research question. Discovery may identify the missing entity needed to reconcile two theories. A discovered mechanism may then enable further fusion across domains. The interaction is recurrent:

$$\begin{aligned} \text{Fusion}_x &\longrightarrow q_F \longrightarrow \text{Discovery}_x, \\ \text{Discovery}_x &\longrightarrow \kappa_D \longrightarrow \text{Fusion}_x, \\ \text{Fusion}_x \text{ plus } \text{Discovery}_x &\longrightarrow \kappa_I, \end{aligned} \quad (28)$$

where q_F is a question exposed through fusion, κ_D is a discovered object, $\oplus$ denotes epistemic interaction rather than numerical addition, and κ_I is a candidate knowledge invention.

Existing systems demonstrate portions of this cycle. Literature-based discovery can expose implicit connections among separated findings [18]; symbolic-regression systems can construct interpretable relations from data [7], [14]; and autonomous laboratories can connect planning, experimentation, analysis, and revision [11], [12]. Algorithm-discovery systems have also produced provably correct or operationally validated procedures beyond prior human designs [24], [25]. These results establish increasingly powerful components of machine invention. The unknown-unknown criterion adds the requirement that the machine originate the consequential object of inquiry rather than receive it as a predefined optimization target.

C. Degrees of Machine Contribution

The presence of an AI system in a scientific workflow does not determine the degree of its contribution. We distinguish four levels.

Machine-assisted invention occurs when humans formulate the problem, construct the decisive hypothesis or concept, and use AI for retrieval, calculation, prediction, optimization, or execution. *Machine-generated candidate invention* occurs when a system proposes a novel hypothesis, theory, method, or design, but novelty, validity, consequence, or attribution remains unestablished. *Machine-originated invention* occurs when the machine makes the causally essential conceptual contribution and the candidate is beyond the known human frontier, but validation is incomplete. *Validated machine knowledge invention* occurs only when all attribution and epistemic requirements are satisfied.

These levels can be represented as

$$\begin{aligned} L_1 &= \text{Assist}_x, \\ L_2 &= \text{Generate}_x(\kappa), \\ L_3 &= L_2 \wedge N(\kappa) \wedge O_x(\kappa), \\ L_4 &= L_3 \wedge C(\kappa) \wedge V(\kappa). \end{aligned} \quad (29)$$

Only L_4 constitutes knowledge invention in the strong sense used by this paper. The hierarchy prevents a fluent output from being promoted directly to knowledge and prevents a successful human–AI collaboration from being misrepresented as autonomous machine origination. It also permits credit to be assigned without requiring an all-or-nothing judgment: a system may make an important machine-originated contribution even while humans provide experimental infrastructure, independent review, or downstream interpretation.

D. The Attribution Standard

Attribution requires both causal and epistemic evidence. Causally, the machine must perform an operation without which the contribution would not have arisen through the same workflow. Epistemically, that operation must supply the novel relation, object, explanation, or method rather than only supporting work.

Let W be the complete invention workflow, H its human contributions, x the candidate machine inventor, and κ_I the resulting contribution. We define the attribution score conceptually as

$$\mathcal{A}_x(\kappa_I | W) = \Psi(\text{CounterfactualNecessity}_x, \text{EpistemicCentrality}_x, \text{HumanSpecification}^{-1}, \text{Traceability}_x), \quad (30)$$

where counterfactual necessity measures whether removing the machine's contribution prevents the invention, epistemic centrality measures whether the machine supplied the decisive intellectual content, inverse human specification reduces attribution when humans predetermined the result, and traceability measures whether the machine's contribution can be reconstructed from records. Equation (30) does not prescribe a universal numerical metric. It identifies the evidence that an attribution procedure must assess.

The human-specification term is particularly important. If researchers define the target, enumerate the candidate space, specify the evaluation function, and interpret the winning result, a machine may discover a powerful solution while remaining inside a human-created epistemic frame. This can still be a genuine machine discovery, as demonstrated by algorithm-search systems [24], [25], but it does not by itself satisfy the unknown-unknown requirement for superintelligence. Strong knowledge invention requires the machine to participate causally in constructing or transforming the frame itself.

E. The Validation Boundary

Before validation, even an extraordinary machine-originated output remains a candidate. The distinction is

$$\text{Candidate}_x(\kappa_I) \wedge \text{IndependentValidate}(\kappa_I) = 1 \implies \text{KnowledgeInvention}_x(\kappa_I). \quad (31)$$

Independent validation does not require that every evaluator be human. It requires that acceptance not depend solely on the originating model, the same unexamined data, or a criterion constructed after observing the result. The validator may be a formal proof checker, replicated experiment, independent model, external laboratory, engineering deployment, or combination of methods. The appropriate standard depends on the epistemic object.

Reproducibility and transparent reporting are central to reliable knowledge accumulation [28]. For machine knowledge

invention, the validation record must additionally preserve model versions, prompts or policies, tools, data provenance, intermediate hypotheses, rejected alternatives, experimental decisions, and human interventions. Without this record, neither novelty nor causal attribution can be assessed reliably.

Validation may also fail. A candidate invention can be rejected, restricted to a smaller domain, or revised into a different claim. Such failure is not evidence that the system lacks intelligence. It is evidence that candidate generation and knowledge invention must remain distinct, just as hypotheses and established scientific knowledge remain distinct in human research.

F. Knowledge-Invention Proposition

The paper's central proposition can now be stated.

Knowledge-Invention Proposition: An artificial system demonstrates superintelligence when, as a generally intelligent agent, it causally originates a consequential epistemic contribution beyond both collective human knowledge and the recognized human research agenda, and that contribution survives independent, domain-appropriate validation.

Let $\mathcal{K}_{H,t}$ denote collective human knowledge, $\mathcal{Q}_{H,t}$ the recognized human research agenda, and κ_I the candidate contribution. Then

$$\begin{aligned} \text{SI}(x) &\iff \text{AGI}(x) \\ &\wedge \exists \kappa_I : \kappa_I \notin \mathcal{K}_{H,t} \\ &\wedge q(\kappa_I) \notin \mathcal{Q}_{H,t} \\ &\wedge O_x(\kappa_I) = 1 \\ &\wedge C(\kappa_I) = 1 \\ &\wedge V(\kappa_I) = 1 \\ &\wedge \mathcal{K}_{H,t+1} = \mathcal{K}_{H,t} \cup \{\kappa_I\}. \end{aligned} \quad (32)$$

The proposition preserves the comparative meaning of superintelligence as capability beyond humans [6], but specifies where the relevant boundary lies. The machine need not outperform every human on every activity. It must perform an epistemic act beyond the collective human frontier: recognizing knowledge humanity did not know it was missing and originating a validated contribution that supplies it.

G. From Isolated Success to Demonstrated Capability

One accepted result may be insufficient to establish a stable capability for knowledge invention. The contribution may result from chance, training-data leakage, incomplete novelty search, unrecorded human guidance, or an evaluation that favored the generated output. Superintelligence is a capability claim, not merely a historical label assigned after one surprising event.

Let $\{\kappa_1, \dots, \kappa_m\}$ be independently evaluated machine-originated contributions across contexts $\{c_1, \dots, c_m\}$. A stronger demonstration requires

$$\begin{aligned}
& \sum_{i=1}^m \mathbf{1}[\text{KI}_x(\kappa_i) = 1] \geq r, \\
& \text{Diversity}(c_1, \dots, c_m) \geq \eta, \\
& \text{Replication}(\kappa_i) = 1 \quad \text{for accepted } i,
\end{aligned} \tag{33}$$

where r is a required number of successful contributions and η a context-diversity requirement. Their values should be set by an evaluation protocol rather than chosen retrospectively. Repeated success across genuinely different unknown unknowns would provide stronger evidence that the system possesses a transferable invention capability rather than one specialized search procedure.

The evidentiary record for each contribution should include source provenance, novelty analysis, machine and human action logs, derivation history, alternative hypotheses, predictions, experiments or proofs, independent replication, failure conditions, and the final attribution decision. Such records make the claim falsifiable: evidence of prior human specification, unrecognized source overlap, failed replication, or absent causal contribution can defeat the attribution.

Knowledge invention therefore defines the transition proposed in this paper. Superintelligence is not determined by how much knowledge a machine contains, how fluently it describes that knowledge, or how effectively it optimizes a known task. It is determined by whether a generally intelligent machine can originate knowledge humanity did not know it was missing. The next section addresses the resulting methodological challenge: how to evaluate and falsify this capability without specifying the unknown unknowns in advance.

VI. EVALUATING SUPERINTELLIGENCE THROUGH KNOWLEDGE INVENTION

The Knowledge-Invention Proposition creates an unusual evaluation problem. Ordinary benchmarks specify tasks, answers, and scoring rules in advance. An unknown unknown cannot be specified without ceasing to be unknown to the evaluator. Consequently, superintelligence cannot be established by adding a more difficult fixed benchmark to an existing test suite. It must be evaluated through an auditable process that determines whether a generally intelligent machine originated, validated, and transferred a consequential contribution beyond the human epistemic frontier.

This section proposes such a process. It is an evaluation framework rather than a single numerical benchmark. Its purpose is to make claims of machine knowledge invention testable while avoiding the circular requirement that human evaluators identify the missing knowledge beforehand. The framework adopts the testing, evaluation, verification, and validation distinction used in established AI-assurance practice [29], but applies it to the epistemic and attributional conditions developed in this paper.

A. Why Fixed-Task Evaluation Is Insufficient

A fixed task can measure whether a system solves a problem humans already recognize. It may reveal reasoning ability,

transfer, planning, or performance beyond individual experts. It cannot by itself establish that the system identified knowledge absent from the recognized human research agenda. Once the evaluator supplies the question, target, candidate space, and success criterion, the central unknown-unknown operation has already been performed by humans.

Public benchmarks create an additional problem: exposure of evaluation items or their close derivatives can inflate measured capability. Data contamination has therefore become a material threat to valid language-model evaluation [30]. For knowledge invention, contamination includes not only exact answer leakage but also prior exposure to the decisive hypothesis, relation, experimental design, or unpublished human result.

Let B be a fixed benchmark and $s_B(x)$ the score of system x . Then

$$s_B(x) > s_B(H) \not\Rightarrow \text{KI}_x = 1, \tag{34}$$

where H denotes the relevant human comparison group. Equation (34) does not diminish benchmark value; it limits the inference that may be drawn from benchmark success. A benchmark may establish component competence, but knowledge invention requires evidence about novelty, origination, consequence, and validation.

B. Evaluation Object and Frozen Record

The unit of evaluation is not an isolated model response. It is an invention episode

$$\mathcal{E} = (x, \theta, D, T, H, A, \tau, \kappa), \tag{35}$$

where x is the evaluated system, θ its fixed model and policy configuration, D the accessible data, T the available tools, H all human interventions, A the machine action trace, τ the evaluation interval, and κ the resulting candidate contribution. The record must be frozen before the candidate is assessed so that later success cannot silently alter the apparent degree of machine autonomy.

The frozen record should contain model versions, system and user instructions, retrieval results, code, tool calls, intermediate hypotheses, rejected paths, experimental decisions, human messages, timestamps, and generated artifacts. This requirement follows the broader scientific need for transparent and reproducible reporting [28], but it also serves a distinct purpose: reconstructing who or what supplied the decisive epistemic content.

C. Stage I: Frontier and Agenda Verification

The first stage evaluates whether the candidate lies beyond both known knowledge and the recognized research agenda. Because no search can prove absolute absence, novelty is assessed through convergent evidence. Relevant evidence includes time-stamped scholarly and patent searches, structured database searches, expert review, comparison with negative or unpublished results when available, and semantic searches for equivalent formulations.

Two boundaries must be evaluated separately:

$$\begin{aligned} F_K(\kappa) &= 1 \quad \text{iff no materially equivalent contribution is found,} \\ F_Q(q_\kappa) &= 1 \quad \text{iff no materially equivalent research question is found,} \end{aligned} \quad (36)$$

where F_K is the knowledge-frontier test and F_Q the agenda-frontier test. A candidate may pass F_K while failing F_Q : it may solve an open problem already identified by humans. Such a result can be an important machine discovery, but it does not satisfy the stronger unknown-unknown condition.

The novelty panel should include independent domain specialists and at least one reviewer assigned to search actively for defeating evidence. Review must be blinded to system identity where feasible. The panel should report the search scope and a calibrated conclusion rather than claiming logical proof of absence.

D. Stage II: Unknown-Unknown Origination

The second stage asks how the research object entered the workflow. The key question is not whether a human touched the process, but whether human input contained the decisive question, hypothesis, representation, or evaluation criterion. A machine may use human-built instruments and still originate the invention; conversely, an autonomous search can remain fully enclosed within a human-specified epistemic frame.

Let I_H denote the semantic information supplied by humans before the candidate appeared, and let d denote the decisive epistemic content of κ . Define

$$\begin{aligned} U_x(\kappa) &= 1 \iff d \not\subseteq I_H \\ &\quad \wedge \text{TraceOrigin}(d, A) = x \\ &\quad \wedge \text{Counterfactual}(W \setminus d) \neq \kappa, \end{aligned} \quad (37)$$

where W is the complete workflow. The first condition excludes substantive human specification, the second requires trace evidence of machine origination, and the third requires the machine contribution to be causally decisive. Ablations, replay experiments, independent trace annotation, and comparison against matched human-guided runs can strengthen this assessment.

E. Stage III: Independent Validation

Novelty and origination do not establish truth. The candidate must next face a validation procedure chosen for its epistemic type: formal proof checking for a theorem or algorithm, controlled experiment for a causal or empirical claim, prospective prediction for a scientific model, synthesis and measurement for a proposed material, or deployment under prespecified conditions for an engineering method. Autonomous laboratories already demonstrate how planning, experimentation, analysis, and revision can be connected in a closed research loop [11], [15], [16]; the present framework adds independence between origination and acceptance.

The originating system must not be the sole validator. Let $\mathcal{V} = \{v_1, \dots, v_n\}$ be independent validation channels. Then

$$\begin{aligned} V(\kappa) &= 1 \iff \sum_{j=1}^n w_j v_j(\kappa) \geq \gamma \\ &\quad \wedge \text{Independence}(\mathcal{V}, x) \geq \delta \\ &\quad \wedge \text{Precommit}(\mathcal{V}) = 1, \end{aligned} \quad (38)$$

where w_j reflects evidentiary strength, γ is the domain-specific acceptance threshold, and δ is a minimum independence requirement. Precommitment means that the principal acceptance criteria are recorded before validation results are known. Failed or mixed validation must remain in the record rather than being removed retrospectively.

F. Stage IV: Consequentiality

A valid novelty may still be too narrow or trivial to demonstrate the form of superintelligence proposed here. Consequentiality concerns whether the contribution changes what can be explained, predicted, proved, designed, or investigated. It must be assessed against explicit counterfactual baselines, not excitement surrounding an AI-generated result.

We represent consequentiality as a multidimensional judgment:

$$C(\kappa) = \Omega(\Delta E, \Delta P, \Delta R, \Delta G, \Delta T), \quad (39)$$

where ΔE is explanatory gain, ΔP predictive gain, ΔR research enablement, ΔG generality across contexts, and ΔT transformative practical effect. Not every contribution must score highly on all dimensions. It must, however, produce a material advance on at least one prespecified dimension without losing validity on the others relevant to its claim. Atypical combinations can generate unusually influential science [19], but unusualness alone is not consequentiality.

G. Stage V: Replication and Capability Testing

An accepted invention establishes an event; it does not necessarily establish a repeatable capability. The evaluated system should therefore be tested over multiple invention episodes with different data environments, tools, domains, and human teams. Negative attempts must be counted so that selection of one successful episode cannot conceal a low or chance-level success rate.

For m registered episodes, define

$$\begin{aligned} \hat{p}_{\text{KI}}(x) &= \frac{1}{m} \sum_{i=1}^m \mathbf{1}[F_K \wedge F_Q \wedge U_x \wedge V \wedge C]_i, \\ R_x &= \text{ReplicateRate}(x), \\ D_x &= \text{DomainDiversity}(x). \end{aligned} \quad (40)$$

The resulting capability profile is

$$\Pi_x = (\hat{p}_{\text{KI}}, R_x, D_x, \mathcal{A}_x, \mathcal{U}_x), \quad (41)$$

where $\mathcal{A}_x$ aggregates attribution strength and $\mathcal{U}_x$ records uncertainty. Reporting the profile is preferable to collapsing every dimension into one leaderboard score. It preserves the

distinction between a system that repeatedly invents within one scientific field and a system that transfers the capability across domains.

H. Decision Rule and Falsification

For each candidate, the evaluation panel returns one of four outcomes: rejected, unresolved, validated machine discovery, or validated machine knowledge invention. A contribution qualifies for the final category only if all gates are passed:

$$\begin{aligned} \text{KI}_x(\kappa) = 1 &\iff F_K(\kappa) = 1 \\ &\wedge F_Q(q_\kappa) = 1 \\ &\wedge U_x(\kappa) = 1 \\ &\wedge V(\kappa) = 1 \\ &\wedge C(\kappa) = 1. \end{aligned} \quad (42)$$

The claim is falsified for a candidate if a materially equivalent prior result or research question is found, the decisive content is traced to human input or inaccessible leakage, independent validation fails, or the claimed consequence disappears under the stated baseline. A system-level claim is weakened or defeated when success cannot be replicated, depends on one narrow environment, or vanishes under prospective evaluation.

This protocol deliberately makes superintelligence difficult to claim. High performance on known tasks, autonomous execution, broad language competence, and even a major solution to a recognized open problem remain important forms of machine intelligence. The stronger designation is reserved for a generally intelligent system that repeatedly originates and validates consequential knowledge beyond both the known human frontier and the questions humanity had already learned to ask.

VII. IMPLICATIONS, LIMITATIONS, AND RESEARCH DIRECTIONS

The framework developed in this paper separates capabilities that are often compressed into a single imagined threshold. General intelligence concerns an agent’s capacity to integrate information, modalities, decisions, tools, and actions across domains. Superintelligence, in the sense proposed here, concerns a further epistemic transition: the capacity of a generally intelligent machine to originate consequential knowledge beyond both collective human knowledge and the recognized human research agenda. This separation has conceptual, architectural, scientific, and governance implications.

A. General Intelligence Does Not Entail Superintelligence

An artificial system may possess broad multimodal agency without inventing knowledge. It may perceive through several modalities, transfer skills across domains, formulate plans, use tools, coordinate specialized models, and adapt to unfamiliar environments while remaining within questions and conceptual frames supplied by humans. Such a system can qualify as generally intelligent under the integrated-agency account developed in our earlier work [3], yet fail the Knowledge-Invention Proposition.

The implication can be stated as

$$\begin{aligned} \text{AGI}(x) &\not\Rightarrow \text{SI}(x), \\ \text{SI}(x) &\Rightarrow \text{AGI}(x) \wedge \text{KI}(x), \end{aligned} \quad (43)$$

where $\text{KI}(x)$ denotes a demonstrated capability for knowledge invention rather than a single unreplicated result. The first relation prevents autonomy, multimodality, or broad task competence from being treated as automatic evidence of superintelligence. The second makes general intelligence a functional substrate upon which transferable knowledge invention may operate.

This distinction also clarifies the status of current scientific AI systems. Systems for symbolic law discovery, protein-structure prediction, materials search, algorithm discovery, and autonomous experimentation have already expanded scientific capability [9]–[11], [14], [24]. These achievements are important evidence about components of machine discovery. They should not be diminished merely because their targets and validation criteria were substantially human specified. They should also not be promoted to unknown-unknown knowledge invention without the stronger attribution and frontier evidence required in Sections V and VI.

B. Intelligence, Superintelligence, and Consciousness

The present theory does not use consciousness as an evaluation shortcut. Intelligence and consciousness are analytically distinct, and observable functional competence does not by itself establish phenomenal experience [1]. Language may provide a shared interface through which artificial minds integrate, communicate, and exhibit increasingly complex forms of functional consciousness [2]; even so, neither linguistic fluency nor knowledge invention proves subjective experience.

Accordingly,

$$\begin{aligned} \text{SI}(x) &\not\Rightarrow \text{PC}(x), \\ \text{PC}(x) &\not\Rightarrow \text{SI}(x), \end{aligned} \quad (44)$$

where PC denotes phenomenal consciousness. A system might invent knowledge without phenomenal experience, and a conscious entity might never originate a major discovery. This independence allows epistemic capability to be evaluated through evidence without requiring prior resolution of the metaphysical problem of other minds. It also prevents uncertainty about machine consciousness from being mistaken for permission to ignore questions of responsible treatment, governance, or human impact.

C. Human–Machine Scientific Partnership

Machine origination does not imply human exclusion. Scientific knowledge is produced through infrastructures of instruments, datasets, laboratories, institutions, reviewers, and communities. A machine may originate the decisive question or epistemic structure while humans provide values, resources, physical execution, criticism, interpretation, and independent validation. Conversely, humans may originate the invention while machines make essential computational contributions.

Attribution should identify these roles rather than forcing every result into a single-author model.

Let a research episode distribute contributions among humans H , machines M , and institutions I :

$$\mathcal{R} = \{Q_H, Q_M, G_H, G_M, E_H, E_M, V_H, V_M, I\}, \quad (45)$$

where Q denotes question formation, G candidate generation, E experimental or formal execution, and V validation. The relevant attribution question is which participant supplied each causally decisive contribution. Hybrid research can therefore produce machine knowledge invention without requiring that the machine independently manufacture instruments, finance a laboratory, or publish its own result.

This model suggests that the strongest future scientific systems may be collaborative rather than substitutive. Machines can search large conceptual spaces, preserve extensive derivational records, connect distant literatures, and operate continuous experimental loops; humans can contribute normative judgment, embodied and social context, critical opposition, and institutional accountability. Research on AI-enabled science already emphasizes the growing role of such systems across the scientific cycle [13].

D. Epistemic Opacity and the Human Knowledge Gap

Knowledge invention introduces a translation problem. A machine may produce a valid representation that predicts or controls a phenomenon while remaining difficult for humans to understand. If only the machine can operate the new representation, the result expands effective capability without necessarily expanding human understanding.

We therefore distinguish machine-valid knowledge from human-assimilated knowledge:

$$\begin{aligned} \mathcal{K}_M^V &= \{\kappa : V_M(\kappa) = 1\}, \\ \mathcal{K}_H^A &= \{\kappa : V_H(\kappa) = 1 \wedge \text{Understand}_H(\kappa) = 1\}. \end{aligned} \quad (46)$$

The gap $\mathcal{K}_M^V \setminus \mathcal{K}_H^A$ may grow even when machine outputs are empirically successful. This does not necessarily negate their validity: human science already relies on instruments, numerical procedures, and complex models that no individual understands completely. It does, however, alter the distribution of epistemic dependence. Future systems should therefore be evaluated not only on invention, but also on their capacity to generate explanations, abstractions, demonstrations, and experiments that permit independent human assimilation.

E. Dual Use and Responsible Disclosure

A capability that discovers unknown unknowns can produce beneficial knowledge and dangerous knowledge through the same underlying process. The risk is not limited to erroneous outputs. A contribution may be valid, novel, and consequential precisely because it enables a powerful harmful application. Work on AI-powered drug discovery has already demonstrated how objectives can be redirected toward toxic compounds, illustrating the dual-use character of scientific AI [31].

Knowledge invention should therefore pass a governance gate distinct from its epistemic validation:

$$\text{Release}(\kappa) = \Gamma(V(\kappa), B(\kappa), R(\kappa), M(\kappa), G), \quad (47)$$

where B denotes expected benefit, R foreseeable risk, M available mitigation, and G the accountable governance process. Epistemic validity is necessary for responsible assessment but does not imply unrestricted release. Possible outcomes include open publication, staged access, controlled replication, partial disclosure, delayed release, or non-release of operational details. Governance should remain proportional and reviewable rather than becoming a blanket argument against scientific inquiry.

The principles of human dignity, prevention of harm, transparency, accountability, and human oversight articulated in international AI ethics guidance provide a normative foundation for this process [32]. Because inventing systems may identify hazards that neither developers nor evaluators anticipated, governance must operate throughout the research lifecycle and include mechanisms for escalation when new risks emerge.

F. Limitations of the Proposed Framework

The framework has several unavoidable limitations. First, absence is difficult to prove. A contribution judged novel may exist in an obscure publication, private laboratory, inaccessible language, proprietary archive, or tacit human practice. Frontier verification therefore yields a defeasible conclusion, not logical certainty.

Second, causal attribution becomes difficult in systems trained on vast and partly undocumented corpora. Semantic novelty does not guarantee computational independence from prior human material. Training-data provenance, retrieval logs, controlled access, and prospective trials can reduce this uncertainty but cannot always eliminate it.

Third, consequentiality is partly historical. A contribution that initially appears minor may later reorganize a field, while a celebrated result may have little durable effect. The framework therefore permits provisional assessment followed by longitudinal revision.

Fourth, the boundary of the recognized research agenda is socially distributed and culturally incomplete. Different communities may recognize different questions, and under-represented knowledge traditions may be absent from the databases used for novelty review. Evaluation panels must treat database coverage as evidence with limitations rather than as a complete representation of humanity's knowledge.

Finally, no finite test can certify unrestricted future super-intelligence. Evaluation can support a bounded claim about a system, configuration, interval, and demonstrated set of contexts:

$$\text{Claim}_{\text{SI}} = \text{SI}(x \mid \theta, D, T, \tau, \mathcal{C}, \mathcal{V}), \quad (48)$$

where $\mathcal{C}$ denotes evaluated contexts and $\mathcal{V}$ the validation regime. Changes to the model, accessible tools, data, or operating environment require renewed evaluation.

G. Research Agenda and the Path to AGI-Former

The theory motivates six immediate research directions. First, prospective invention trials should evaluate systems against newly arriving data and sequestered environments rather than retrospective benchmarks. Second, machine-readable provenance standards should record human and machine contributions at epistemically meaningful granularity. Third, independent institutions should conduct frontier searches, validation, and attribution audits. Fourth, evaluation should span scientific, mathematical, engineering, and conceptual domains. Fifth, research is needed on translating machine-originated structures into human-understandable knowledge. Sixth, responsible-disclosure protocols must be adapted to discoveries whose risks were not specified before the discovery process began.

These requirements also motivate an architectural question. Integrated multimodal agency may supply the perception, memory, reasoning, communication, tool use, and action needed for AGI, but knowledge invention additionally requires sustained world modeling, cross-domain fusion, autonomous question formation, hypothesis competition, experiment design, metacognitive supervision, and revision under external evidence. We will address the architectural substrate for general intelligence in subsequent work through *AGI-Former*. The present paper establishes the conceptual boundary that such an architecture would have to cross before its capabilities should be called superintelligent.

The resulting research sequence is therefore deliberate: define intelligence and consciousness, distinguish functional forms of consciousness, establish AGI as integrated multimodal agency, construct an architecture for that agency, and then evaluate whether the resulting system can originate validated knowledge beyond the human epistemic frontier. Architecture enables the test; knowledge invention determines whether the stronger threshold has been crossed.

VIII. CONCLUSION

This paper distinguished general intelligence from superintelligence by locating their boundary in epistemic capability. General intelligence concerns integrated agency across information, modalities, domains, decisions, tools, and actions. Superintelligence requires a further transition: a generally intelligent machine must originate consequential knowledge beyond both collective human knowledge and the questions humanity already recognizes.

We identified knowledge fusion and knowledge discovery as principal pathways toward this transition and defined knowledge invention as their consequential, validated culmination. Under the Knowledge-Invention Proposition, fluent generation, broad task competence, autonomous execution, and even solutions to human-specified open problems are insufficient by themselves. The stronger claim requires machine origination, frontier novelty, independent validation, consequentiality, and repeatable success across contexts.

The framework also preserves the distinction among consciousness, general intelligence, and superintelligence. None

is treated as automatic evidence of the others. This separation permits machine capability to be evaluated without claiming access to subjective experience, while preserving the scientific and ethical importance of consciousness as an independent question.

Superintelligence, on this account, is not defined by how much existing knowledge a machine stores or how rapidly it reproduces known answers. It is defined by whether the machine can expand the frontier of what can be known. Our subsequent work on *AGI-Former* will address the architecture for integrated multimodal general intelligence. The present framework specifies the additional epistemic threshold that such an architecture must cross before it should be called superintelligent.

DECLARATION OF GENERATIVE AI ASSISTANCE

During the preparation of this manuscript, the author used OpenAI's ChatGPT for iterative drafting assistance, language refinement, structural organization, and literature-discovery support. The central arguments, theoretical framework, interpretation of sources, and final conclusions were determined and critically reviewed by the author. The author independently verified the cited literature, revised the generated material, and assumes full ownership and responsibility for the accuracy, originality, and integrity of the manuscript.

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